DIODES

Q1. A crystal diode has ……

1. **one pn junction**
2. two pn junctions
3. three pn junctions
4. none of the above

Q2. A crystal diode has forward resistance of the order of ……

1. kΩ
2. **Ω**
3. MΩ
4. none of the above

Q3. If the arrow of crystal diode symbol is positive w.r.t. bar, then diode is …….. biased.

1. **forward**
2. reverse
3. either forward or reverse
4. none of the above

Q4. The reverse current in a diode is of the order of …….

1. kA
2. mA
3. **μA**
4. A

Q5. The forward voltage drop across a silicon diode is about ……

1. 2.5 V
2. 3 V
3. 10 V
4. **0.7 V**

Q6. A crystal diode is used as ……

1. an amplifier
2. **a rectifier**
3. an oscillator
4. a voltage regulator

Q7. The d.c. resistance of a crystal diode is …….. its a.c. resistance

1. the same as
2. more than
3. **less than**
4. none of the above

Q8. An ideal crystal diode is one which behaves as a perfect …….. when forward biased.

1. **conductor**
2. insulator
3. resistance material
4. none of the above

Q9. The ratio of reverse resistance and forward resistance of a germanium crystal diode is about ……

1. 1 : 1
2. 100 : 1
3. 1000 : 1
4. **40,000 : 1**

Q 10. The leakage current in a crystal diode is due to …….

1. **minority carriers**
2. majority carriers
3. junction capacitance
4. none of the above

Q11. If the temperature of a crystal diode increases, then leakage current ……..

1. remains the same
2. decreases
3. **increases**
4. becomes zero

Q12. The PIV rating of a crystal diode is …….. that of equivalent vacuum diode

1. the same as
2. **lower than**
3. more than
4. none of the above

Q13. If the doping level of a crystal diode is increased, the breakdown voltage……

1. remains the same
2. is increased
3. **is decreased**
4. none of the above

Q14. The knee voltage of a crystal diode is approximately equal to ……

1. applied voltage
2. breakdown voltage

3. forward voltage
4. **barrier potential**

Q15. When the graph between current through and voltage across a device is a straight line, the device is referred to as …….

1. **linear**
2. active
3. nonlinear
4. passive

Q16. When the crystal current diode current is large, the bias is ……

1. **forward**
2. inverse
3. poor
4. reverse

Q17. A crystal diode is a …… device

1. **non-linear**
2. bilateral
3. linear
4. none of the above

Q18. A crystal diode utilises …….. characteristic for rectification

1. reverse
2. **forward**
3. forward or reverse
4. none of the above

Q19. When a crystal diode is used as a rectifier, the most important consideration is ……..

1. forward characteristic
2. doping level
3. reverse characteristic
4. **PIC rating**

Q20. If the doping level in a crystal diode is increased, the width of depletion layer……..

1. remains the same
2. is decreased
3. **in increased**
4. none of the above

Q21. A zener diode has ……..

1. **one pn junction**

2. two pn junctions
3. three pn junctions
4. none of the above

Q22. A zener diode is used as …….

1. an amplifier
2. **a voltage regulator**
3. a rectifier
4. a multivibrator

Q23. The doping level in a zener diode is …… that of a crystal diode

1. the same as
2. less than
3. **more than**
4. none of the above

Q24. A zener diode is always …… connected.

1. **reverse**
2. forward
3. either reverse or forward
4. none of the above

Q25. A zener diode utilizes …….. characteristics for its operation.

1. forward
2. **reverse**
3. both forward and reverse
4. none of the above

Q26. In the breakdown region, a zener didoe behaves like a …… source.

1. **constant voltage**
2. constant current
3. constant resistance
4. none of the above

27. A zener diode is destroyed if it……..

1. is forward biased
2. is reverse biased
3. **carrier more than rated current**
4. none of the above

Q28. A series resistance is connected in the zener circuit to……..

1. properly reverse bias the zener
2. **protect the zener**
3. properly forward bias the zener

4. none of the above

Q29. A zener diode is ……. device

1. **a non-linear**
2. a linear
3. an amplifying
4. none of the above

Q30. A zener diode has …….. breakdown voltage

1. undefined
2. **sharp**
3. zero
4. none of the above

Q31. ……. rectifier has the lowest forward resistance

1. **solid state**
2. vacuum tube
3. gas tube
4. none of the above

Q32. Mains a.c. power is converrted into d.c. power for …….

1. lighting purposes
2. heaters
3. **using in electronic equipment**
4. none of the above

Q33. The disadvantage of a half-wave rectifier is that the…….

1. components are expensive
2. diodes must have a higher power rating
3. **output is difficult to filter**
4. none of the above

Q34. If the a.c. input to a half-wave rectifier is an r.m.s value of 400/√2 volts, then diode PIV rating is …….

1. 400/√2 V
2. **400 V**
3. 400 x √2 V
4. none of the above

Q35. The ripple factor of a half-wave rectifier is ……

1. 2
2. **1.21**
3. 2.5
4. 0.48

Q36. There is a need of transformer for ……..

1. half-wave rectifier
2. **centre-tap full-wave rectifier**
3. bridge full-wave rectifier
4. none of the above

Q37. The PIV rating of each diode in a bridge rectifier is …… that of the equivalent centre-tap rectifier

1. **one-half**
2. the same as
3. twice
4. four times

Q38. For the same secondary voltage, the output voltage from a centre-tap rectifier is …… than that of bridge rectifier

1. twice
2. thrice
3. **four time**
4. **one-half**

Q39. If the PIV rating of a diode is exceeded, ……

1. the diode conducts poorly
2. **the diode is destroyed**
3. the diode behaves like a zener diode
4. none of the above

Q40. A 10 V power supply would use ……. as filter capacitor.

1. paper capacitor
2. mica capacitor
3. **electrolytic capacitor**
4. air capacitor

Q41. A 1,000 V power supply would use …….. as a filter capacitor

1. **paper capacitor**
2. air capacitor
3. mica capacitor
4. electrolytic capacitor

Q42. The …….. filter circuit results in the best voltage regulation

1. **choke input**
2. capacitor input
3. resistance input
4. none of the above

Q43. A half-wave rectifier has an input voltage of 240 V r.m.s. If the step-down transformer has a turns ratio of 8:1, what is the peak load voltage? Ignore diode drop.

1. 27.5 V
2. 86.5 V
3. 30 V
4. **42.5 V**

Q44. The maximum efficiency of a half-wave rectifier is …….

1. **40.6 %**
2. 81.2 %
3. 50 %
4. 25 %

Q45. The most widely used rectifier is …….

1. half-wave rectifier
2. centre-tap full-wave rectifier
3. **bridge full-wave rectifier**
4. none of the above

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